Experiment No. 3	
SE (CSE)	ROLL NO :
Date of Implementation:	
Aim: Implement Data Pre-processing techniques in Python	
Programming Language Used : Python	
Upon completion of this experiment, students will be able to CO2 : Apply various analysis and visualization techniques using modern tools on real-world data	

Indicator	Poor	Average	Good
Timeline Maintains submission deadline (2)	Submission not done (0)	One or More than One week late (1)	Maintains deadline (2)
Completion and Organization (2)	Many mistakes in documents (0)	Document is just acceptable (1)	Completed whole document and neatly organized (2)
Program Performance (2)	Could not perform at all (0)	Implemented few parts (1)	Full implementation (2)
Knowledge In depth knowledge of the Experiment (4)	Unable to answer questions(0)	Unable to answer few questions (1-2)	Able to answer all questions with proper explanation (3-4)

Assessment Marks :

Timeline	
Completion and Organization	
Program Performance	
Knowledge	

TOTAL :

EXPERIMENT	***3***
Aim	Implement Data Preprocessing techniques in Python
Theory	Data Cleaning Data cleaning is the process of identifying and correcting errors, inconsistencies, and inaccuracies present in raw data. Data Transformation Data transformation converts cleaned data into a suitable format for analysis and modeling. It involves changing data types, scaling numerical values, and encoding categorical variables. Data Integration Data integration combines data from multiple sources into a single, consistent dataset. It is commonly required when data is collected from different systems or files. Data Summarization Data summarization provides a compact representation of data using statistical measures. Measures such as mean, median, standard deviation, and group-wise aggregates help in understanding central tendency and variability. Feature Engineering Feature engineering is the process of creating new meaningful features from existing data to improve analytical or predictive performance.
Implementatio n	1. Handle Missing Values i. Replace missing Age with median age ii. Replace missing Income with mean income iii. Replace missing PurchaseAmount with 0 2. Correct Invalid Values Remove records where Age < 0 or Age > 100 3. Standardize Categorical Values i. Convert Gender values to only Male and Female ii. Convert ProductCategory to lowercase 4. Remove Duplicate Records Remove duplicate rows based on CustomerID 5. Perform equal width binning on the Income attribute. i. Divide the Income attribute into 4 equal-width bins. ii. Assign appropriate labels to each bin (Very Low, Low, Medium, High). iii. Create a new feature named Income_EqualWidth. iv. Plot a histogram showing the distribution of customers across the equal-width bins. 6. Perform Equal Frequency Binning i. Divide the Income attribute into 4 equal-frequency bins. ii. Assign labels (Q1, Q2, Q3, Q4). iii. Create a new feature named Income_EqualFrequency. Plot a bar chart showing the number of customers in each bin 7. Outlier detection i. Compute Z-scores for PurchaseAmount.

	ii. Identify records with Z-score > 3 or < -3. iii. Display the outlier records. iv. Remove outliers and recompute the mean. 8. Data Type Conversion Convert Income from string to float 9. Normalization Normalize Age and Income using Min–Max scaling 10. Dataset Integration Perform inner join on CustomerID (Drop orphan purchase records automatically) 11. Feature Creation Create a new feature HighSpender (Mark 1 if PurchaseAmount $>$ average, else 0) 12. OLAP operations i. Perform slice operation to get total and average purchase amount for Female customers only. ii. Perform dice operation for: Gender = Male, ProductCategory = Electronics. Display number of purchases and average PurchaseAmount iii. Perform roll-up operation to compute total purchase amount by Gender. iv. Perform drill down operation. Analyze purchase amount by Gender and ProductCategory. 13. Apply one hot encoding to gender column. Add encoded columns to data set. Drop original gender column
Conclusion	This experiment demonstrates the importance of preprocessing steps in the data analytics pipeline. Properly cleaned and processed data significantly improves the quality and reliability of analytical and predictive models.
Post Lab Questions:	1. What is the difference between pivot_table command and group by command? 2. What are orphan records? 3. What is dummy variable trap?